Classification of Legal Text

Machine Learning mini-project
Team 9 / Problem 5

Sai Sathvik Gullipalli / PES2UG24CS430
Priyanka Mallikarjuna Reddy / PES2UG24CS378

Legal opinion classification

Input

A full English US Supreme Court opinion

Output

One broad issue area, such as Criminal Procedure or Civil Rights

Research question

How do lexical features, topics, and document embeddings compare?

Dataset and category imbalance

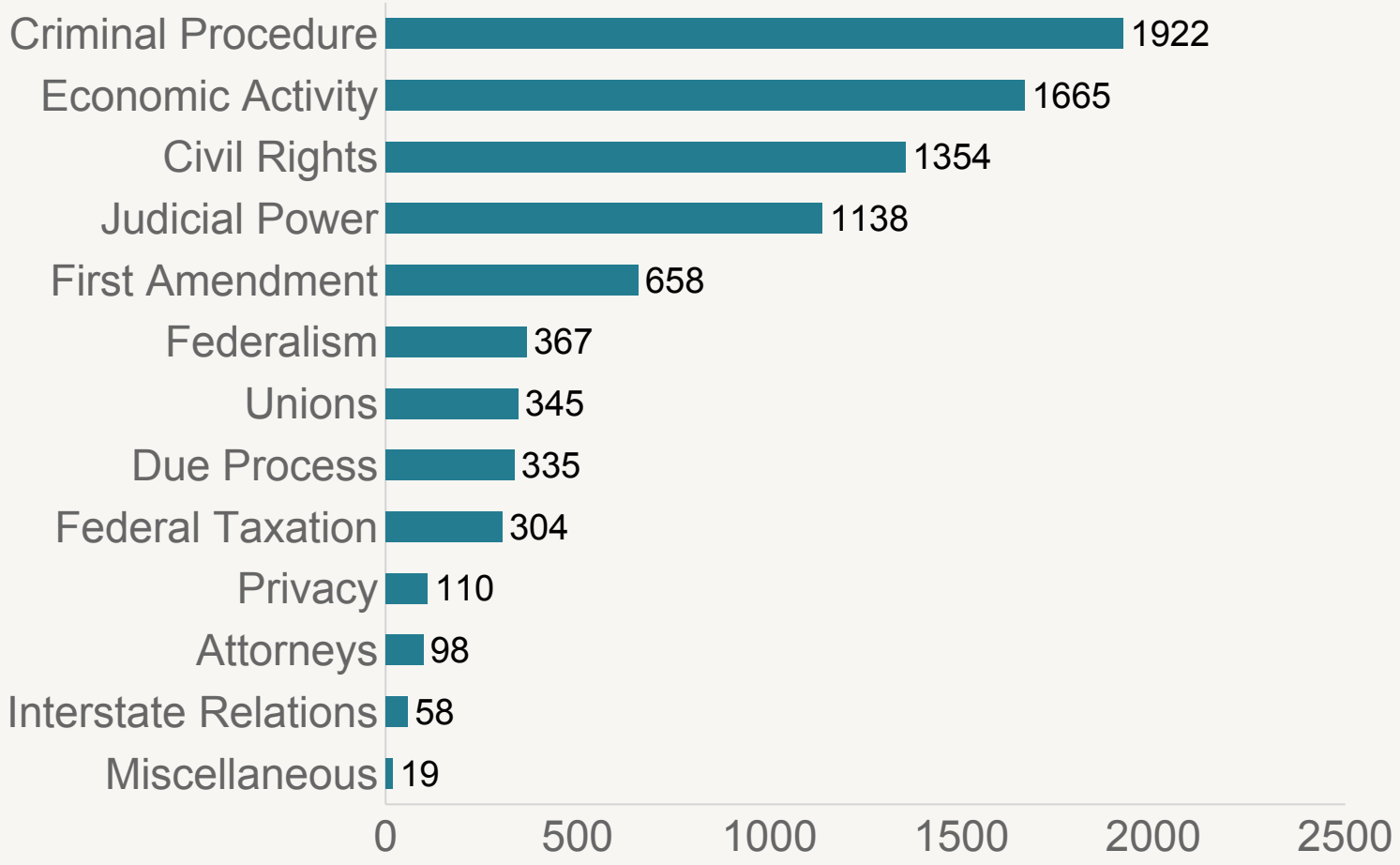
8,419

original opinions

8,373

13 supported issue areas

1946–2016
Private Action excluded: one case
46 records excluded overall



Preparation and evaluation pipeline

Stage	Operation
Clean and group	Link shared citations and normalized text hashes
Split once	5,023 train / 1,675 validation / 1,675 test
Learn representations	Fit vocabularies, LDA, and Doc2Vec on training only
Select and freeze	Validation macro-F1, then accuracy and latency
Evaluate	One held-out comparison of frozen models

Seed 42. Case groups never cross partitions.

Three representations, one classifier

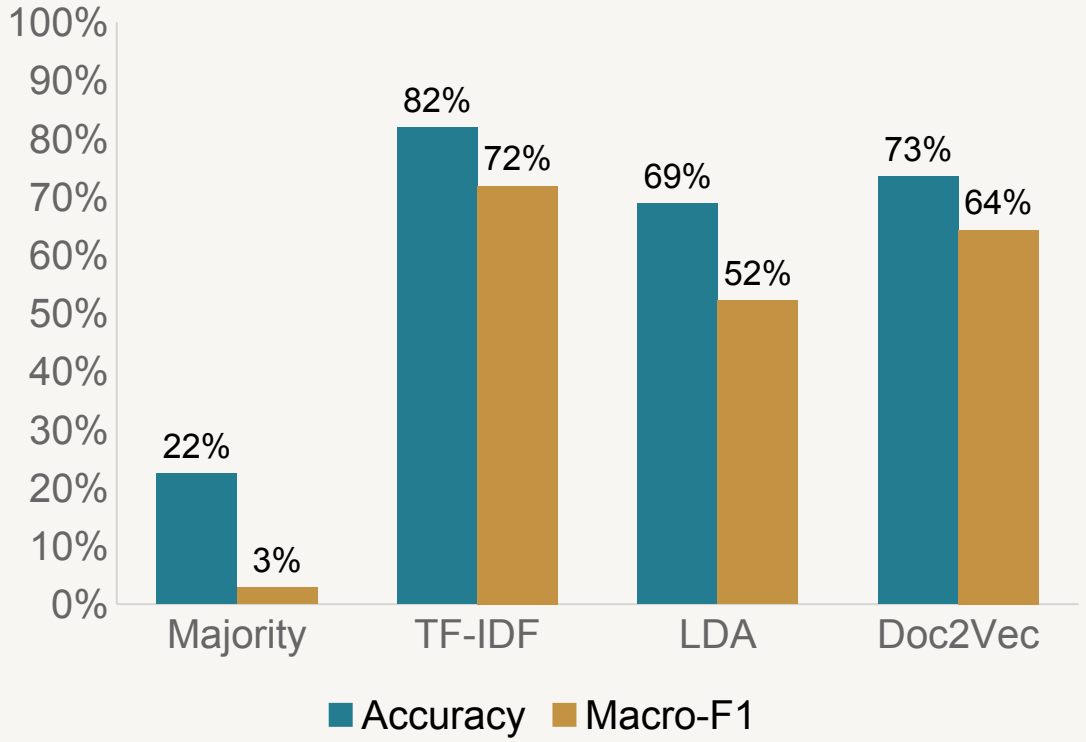
Representation	Features	Training choice
TF-IDF	Unigrams and bigrams	Up to 30,000 features
LDA	Topic proportions	20 or 40 topics, count input
Doc2Vec	Dense document vectors	100 dimensions, 30 epochs

Logistic Regression: C = 0.1, 1, 10

LDA uses counts. Doc2Vec uses inferred vectors consistently across all partitions.

Held-out model comparison

Model	Accuracy	Macro-F1
Majority baseline	0.224	0.028
TF-IDF + LR	0.819	0.719
LDA + LR	0.688	0.522
Doc2Vec + LR	0.735	0.643



Validation-selected model: TF-IDF + LR

Macro-F1 gives each category equal weight. Test scores never select the model.

Saved-model notebook demonstration

Example: Criminal Procedure

TF-IDF prediction: Criminal Procedure

Top category	Model probability
Criminal Procedure	92.2%
Civil Rights	2.9%
Due Process	2.5%

Similar training opinions

MARINO v. RAGEN, WARDEN

MCNEAL v. CULVER, STATE PRISON CUSTODIAN

UVEGES v. PENNSYLVANIA

No retraining. Probabilities are uncalibrated. Similarity does not establish legal authority.

Conclusions and limitations

TF-IDF + LR reaches 81.9% test accuracy

Our comparison supports classification and qualitative opinion retrieval on this historical corpus.

Future-case generalization needs a temporal evaluation.
Small categories and source-label noise limit reliability.
Short passages differ from full opinions.
The US corpus does not establish performance on Indian law.